



RESEARCH

# GPU-parallelized SPH solver for accurate hypervelocity impact simulation of shaped charge jet penetration in concrete structures

Jin Hyun Kim · Hee Sang Yoo · Young Beom Jo · Eung Soo Kim

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**Abstract** Accurately analyzing local failure areas, such as penetration or perforation in concrete structures under extreme conditions, such as those caused by shaped charge jet penetration, is of paramount importance for ensuring structural integrity and safety. This study addresses these challenging problems by developing a GPU-parallelized smoothed particle hydrodynamics solver, which incorporates advanced multiphase SPH models, complex constitutive models, and equations of state (EOS) for metal and concrete materials. Enhanced with variable smoothing lengths, this solver improves the accuracy of simulations. Numerical simulations of high-velocity impacts (HVI) on metal and concrete materials were conducted to validate the solver's capability and precision. The simulations confirmed that shock waves propagate according to material properties, with stable pressure fields and logically coherent crack formations.

Comparative analysis with experimental observations demonstrated improved accuracy, with relative errors for depth of penetration (DOP) and average borehole diameter under 5%. Furthermore, parametric tests examining variations in shaped charge geometry and concrete compressive strength showed reasonable variations in crater morphology. These results indicate that the developed SPH solver effectively addresses a wide range of shaped charge jet collision problems and presents a promising alternative to experimental methods for extreme conditions.

**Keywords** SPH · GPU · Elasto-viscoplastic · Shaped charge · Concrete

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## 1 Introduction

Concrete is a composite material that plays a pivotal role in both civilian and military infrastructure, including building and civil infrastructures as well as bunkers, fortifications, and protective barriers. Its critical function extends to containment structures in various power plants, high-level radioactive waste repositories, wastewater treatment systems, and flood defense systems, where it serves as an essential barrier in each application. Therefore, the importance of the structural integrity of these concrete installations is undisputable. However, numerous reports and studies

(NEI 2011; US NRC 2011; Wilt et al. 2011), indicate these structures are vulnerable to displacement or collapse when subjected to the penetration of high-impact weapons. Such impacts typically generate high pressures at the point of contact, leading to various structural failures including craters, tunnels, spallation, and cracks. These findings highlight the significant challenges in maintaining the integrity of concrete structures under extreme conditions, as further emphasized by recent research (Wang et al. 2021).

Shaped charges, which are explosives designed to focus blast energy into a high-velocity metal jet, represent one of these severe loading scenarios. The physics behind this process involves complex interactions of shock waves, material deformation, and high strain rates. This mechanism of shaped charges allows for the deep penetration of target materials, making shaped charges a subject of extensive study for their effectiveness in overcoming the resistance of physical barriers (Xu et al. 2017). Due to their powerful penetrating capabilities, shaped charges have found widespread application in evaluating the integrity of metal (Xi et al. 2012; Guo et al. 2016) and concrete target (Esteban et al. 2015; Zhu et al. 2018; Moo et al. 2022).

Numerous experiments have been conducted to validate the integrity of concrete structures under extreme impact. For instance, perforation experiments have been performed with concretes of varying unconfined compressive strengths (48 and 140 MPa), comparing the residual velocities of projectiles (Hanchak et al. 1992). Spherical fragments fired at concrete blocks using explosives at approximately 1,650 m/s have been used to analyze global macro-cracking and material property changes (Leppänen et al. 2005). Additionally, mushroom-shaped projectiles launched at concrete plates at velocities ranging from 100 to 500 m/s have been employed to evaluate local damage (Beppu et al. 2008). However, conducting physical experiments with extreme loading on concrete containment is limited by their complexity, risk, and high cost. As a result, numerical simulation has emerged as a valuable tool for investigating the behavior of materials and structures under extreme loading conditions, providing a more cost-effective and manageable alternative to traditional experiments.

The finite element method (FEM) is a popular choice for numerical simulations of impact problems,

favoring its ability to deliver precise stress and strain analyses under dynamic loads, based on its robustness and established maturity. It is recognized within the engineering community as a reliable method for tackling structural challenges. Recently, FEM has been extensively applied to investigate high-velocity projectile impacts on concrete target, including shaped charge jets (Elshenawy et al. 2018; Simons et al. 2019; Khan and Iqbal 2022; Khan and Sharma 2022). Nevertheless, FEM, being a grid-based method, requires the mesh to be established in the initial domain. This necessitates additional model applications such as remeshing (Cremonesi et al. 2020) and handling nonlinearity treatments (Pascon 2019) when large deformations or fractures extend outside the predefined mesh, increasing the complexity of the numerical treatment. Specifically, high-velocity impacts can lead to severe mesh distortion, reduced accuracy, and potential numerical instability. These issues necessitate sophisticated techniques to maintain mesh quality and stability, which can be computationally intensive and challenging to implement effectively.

Particle-based methods, such as smoothed particle hydrodynamics (SPH), have increasingly gained attention as viable alternatives for numerical simulations due to their inherent advantages in managing complex nonlinearities. These methods do not rely on a predefined grid, making them inherently more flexible in handling large deformations, fractures, and material separation, which are common in nonlinear high-velocity impact scenarios. The absence of a mesh eliminates issues related to mesh distortion and remeshing, thereby enhancing the accuracy and stability of the simulations. Initially, SPH and other particle-based methods were less favored due to their lower computational efficiency compared to grid-based methods. However, advancements in computing technology, particularly the development of graphics processing units (GPUs), have significantly enhanced their computational capabilities. This improvement has spurred increased interest in their application including complex structural analyses (Frissane et al. 2019; Lu et al. 2023). In this evolving context, SPH is now increasingly applied in simulations assessing the concrete response to high-velocity projectiles (Lu and Sonoda 2020; Zhang et al. 2020; Karmakar and Shaw 2021).

Conventional GPU-SPH schemes for extreme load impact have been hampered by two fundamental shortcomings. One is that the fixed smoothing length adopted for memory coalesced kernels leads to density drift and poor shock fidelity. The other is that material-specific particle sets require artificial interface treatments, preventing a fully coupled multiphase description. To overcome these constraints, this study developed a GPU-parallelized SPH solver to accurately analyze local failure areas, such as penetration or perforation, in various materials, particularly concrete structures under extreme conditions. This solver employs a variable smoothing length enhanced with an  $\Omega$ -correction term, eliminating the mass drift while retaining consistency and couples it to an integrated mass conservation equation that remains robust across large density contrasts. Appropriate equations of state and constitutive models store material parameters on each particle, allowing metals and concrete to interact seamlessly within a single particle ensemble, thereby enabling multi physics problems to be solved within a unified computational framework. GPU parallelization drastically reduces computational time while efficiently managing the very large number of particles required for high-fidelity failure analysis. In this study, the phenomena of a shaped charge jet penetration into concrete cylinder were simulated, variables such as depth of penetration and crater diameter that can measure local failure were compared, and the degree of damage to the concrete cylinder was analyzed.

This paper is organized as follows: Sect. 2 provides an overview of the SPH method and presented the various models used in this research. Section 3 performed the numerical validation problem with high velocity impact on metal and concrete material to validate the material models. Section 4 conducted simulations of shaped charge jet penetration into concrete cylinder, including a comparison with experimental data and a parametric study to identify the cases with the most significant penetration. Finally, Sect. 5 concludes the paper, summarizing the findings, and providing recommendations for future research in this field.

## 2 Methodology

In this section, the governing equations are articulated and their discretized versions are explicitly presented and explained, including mass and momentum conservation, as well as the constitutive equation. Additionally, various Equation of State models and constitutive models utilized to deal with various materials in this study are detailed. The overall computing algorithm employed in the analysis is also outlined.

### 2.1 Governing equations

The governing equations for continuum materials in a Lagrangian frame consist of the conservation of mass, momentum, and energy, expressed as follows:

$$\frac{D\rho}{Dt} = -\rho \frac{\partial v^\beta}{\partial x^\beta} \quad (1)$$

$$\frac{Dv^\alpha}{Dt} = \frac{1}{\rho} \frac{\partial \sigma^{\alpha\beta}}{\partial x^\beta} \quad (2)$$

$$\frac{De}{Dt} = \frac{\sigma^{\alpha\beta}}{\rho} \frac{\partial v^\alpha}{\partial x^\beta} \quad (3)$$

where  $\rho$  is the density,  $e$  is the internal energy,  $v^\alpha$  is the velocity component,  $x^\alpha$  is the spatial coordinate and  $\sigma^{\alpha\beta}$  is the total stress tensor, and  $D/Dt$  is the Lagrangian derivative.  $\alpha$  and  $\beta$  are tensor indices that range from 1 to the number of spatial dimensions, typically 3 for three-dimensional spaces.

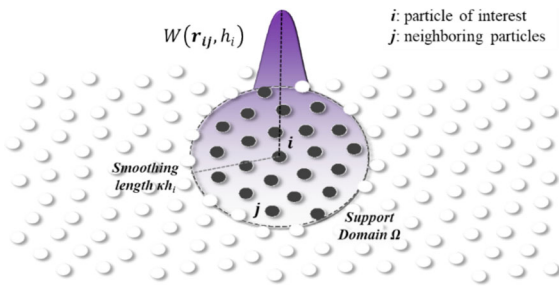
$$\sigma_{\alpha\beta} = -p\delta_{\alpha\beta} + \tau_{\alpha\beta} \quad (4)$$

Isotropic pressure can be computed from the equation of state (EOS), while the shear stress tensor is determined using a constitutive model.

### 2.2 SPH governing equations

In the SPH method, a field variable is approximated as a discretized integration over the neighboring particles as in Fig. 1. This approximation can be expressed mathematically as follows:

$$f_i(r) = \sum_j \left( \frac{m_j}{\rho_j} f_j W(r_{ij}, h_i) \right) \quad (5)$$



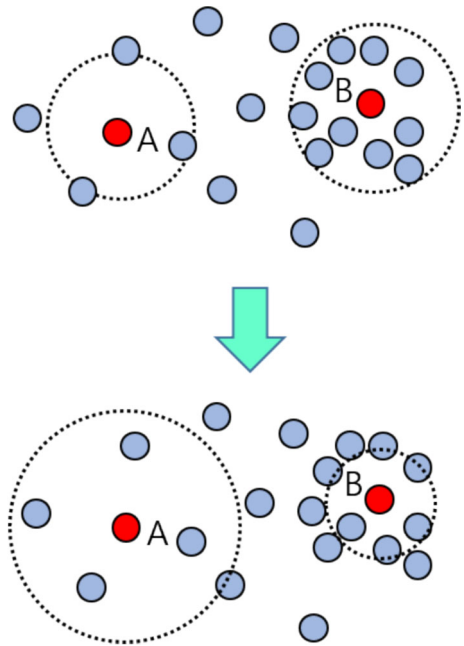
**Fig. 1** Schematic view of SPH approximation

In this equation,  $f_i(r)$  represents the field variable at the position of the center particle  $i$ . The summation runs over all neighboring particles  $j$  that contribute to the SPH approximation. The term  $m_j$  denotes the mass of the neighboring particle  $j$ , and  $\rho_j$  is its density. The value of the field variable at the position of particle  $j$  is represented by  $f_j$ . The function  $W(r_{ij}, h_i)$  is the smoothing kernel, which depends on the distance between particles  $i$  and  $j$  represented by  $r_{ij} := r_i - r_j$  and the smoothing length  $h_i$  associated with particle  $i$ . In this study, we adopt the Wendland C2 kernel as the smoothing kernel, well known for its superior numerical stability (Wendland 1995); notably, it remains free from pairing instability even when a large number of neighbour particles is used (Dehnen and Aly 2012; Yamamoto et al. 2021).

$$W(q) = \begin{cases} C(1-q)^4(1+4q), & \text{if } 0 \leq q < 1 \\ 0, & \text{if } q \geq 1 \end{cases} \quad (6)$$

Here,  $q = r_{ij}/(\kappa h_i)$ ,  $C = 21/(2\pi(\kappa h_i)^3)$  in 3D, and  $\kappa$  as a scale factor (set to 2). The overall support radius of particle can be expressed as  $\kappa h_i$ . The derivation of SPH approximation including the expressions for divergence, gradient, and Laplacian are thoroughly explained in previous literatures (Liu and Liu 2010).

In conventional SPH methods, a constant smoothing length is typically used. However, scenarios such as explosions or impacts may require a variable smoothing length to effectively account for unstable environments. In this case, the smoothing length is adjusted based on variations in density, resulting in distinct smoothing lengths for individual particles. This behavior is illustrated in Fig. 2, where particles in high-density regions have shorter smoothing lengths, while those in low-density regions have wider support domains. In this study, variations in the smoothing



**Fig. 2** Smoothing length adjustment upon neighbor particles condition

length depending on the density were incorporated into the governing equations [For detail, we recommend readers to refer to (Price et al. 2018)]. To manage fluctuations in properties when the smoothing length changes, a term related to the gradient of smoothing length, referred to as term  $\Omega$ , was integrated into the governing equations, which is noted as (Monaghan 2002; Price et al. 2018). The algorithm used to compute the smoothing length for each particle is presented in Fig. 3, which outlines the iterative update procedure based on local density estimates to ensure self-consistency.

$$\Omega_i = 1 - \frac{\partial h_i}{\partial \rho_i} \sum_j m_j \frac{\rho_{0i}}{\rho_{0j}} \frac{\partial W_{ij}(h_i)}{\partial h} \quad (7)$$

$$p_i = f(\rho_i, E_i) \quad (11)$$

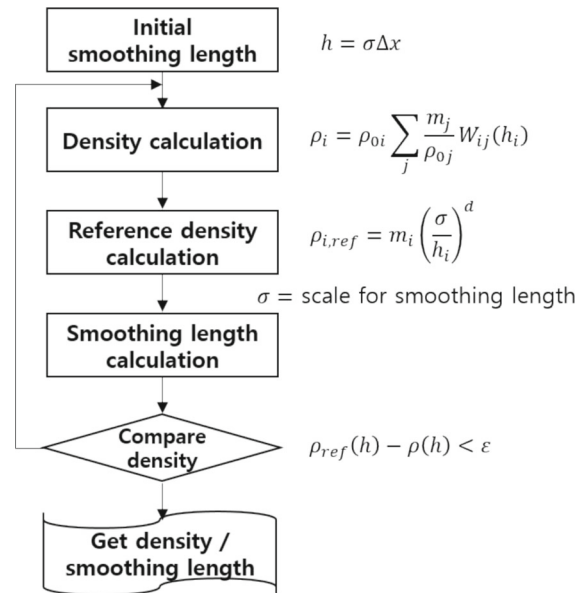
From the original formulation, the term  $\Omega$  was modified to accommodate multiphase flows, taking into account a reference density  $\rho_0$ . Table 1 details the discretized governing equations used in our study. Equation (8) is mass conservation equation which is applicable to both single and multiphase flows (Park et al. 2020). The symmetric forms of the momentum and energy conservation equations are outlined in (9) and (10) respectively. These adjustments allow for an

expanded search range with a larger smoothing length, aligning with Newton's third law. Furthermore, the EOS is specific to each material and is described as a function of density and internal energy.

In order to restrain unphysical particle penetration and minimize computational inaccuracies in the vicinity of the shock wave front, the artificial viscosity is considered (Monaghan 1992):

$$\Pi_{ij} = \begin{cases} \frac{-\alpha c_{ij} \emptyset_{ij} + \beta \emptyset_{ij}^2}{\rho_{ij}}, & v_{ij} \cdot x_{ij} < 0; \\ 0, & v_{ij} \cdot x_{ij} \geq 0; \end{cases} \quad (12)$$

where,  $\emptyset_{ij} = h_{ij} v_{ij} \cdot x_{ij} / (|x_{ij}|^2 + \varphi^2)$ ;  $c_{ij} = (c_i + c_j)/2$ ;  $\rho_{ij} = (\rho_i + \rho_j)/2$ ;  $h_{ij} = (h_i + h_j)/2$ ;  $\nabla_i W_{ij} = \{\nabla_i W_{ij}(h_i) + \nabla_i W_{ij}(h_j)\}/2$ ;  $c$  is the speed of sound;  $\alpha, \beta$  are artificial viscosity coefficients which are set as



**Fig. 3** Smoothing length adjustment algorithm

**Table 1** SPH formulation for governing equation

| Governing equations   | SPH formulation                                                                                                                                                                                                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mass conservation     | $\rho_i = \sum_j \frac{\rho_{0j}}{\rho_{0i}} m_j W(h_i)$ (8)                                                                                                                                                                                                                              |
| Momentum conservation | $\frac{dv_i}{dt} = \sum_j m_j \left[ \frac{\sigma_i}{\Omega_i \rho_i^2} \nabla_i W_{ij}(h_i) + \frac{\sigma_j}{\Omega_j \rho_j^2} \nabla_i W_{ij}(h_j) - \Pi_{ij} \cdot \nabla_i W_{ij} \right]$ (9)                                                                                      |
| Energy conservation   | $\frac{dE_i}{dt} = \frac{1}{2} \sum_j m_j \bar{v}_{ij} \left[ \frac{p_i}{\Omega_i \rho_i^2} \nabla_i W_{ij}(h_i) + \frac{p_j}{\Omega_j \rho_j^2} \nabla_i W_{ij}(h_j) + \Pi_{ij} \cdot \nabla_i W_{ij} \right] + \frac{\tau_{\alpha\beta i} \dot{\epsilon}_{\alpha\beta i}}{\rho_i}$ (10) |

1.5 in this study and  $\varphi = 0.1h_{ij}$  is included to prevent numerical instability when two particles are overlapping.

### 2.3 Elasto-viscoplastic constitutive model for SPH

The elasto-viscoplastic constitutive model is employed to simulate solid impacts involving various materials. The Jaumann stress rate is commonly used in this context, described by the following (13):

$$\frac{d\tau_{\alpha\beta}}{dt} = 2G \left( \dot{\epsilon}_{\alpha\beta} - \frac{1}{3} \delta_{\alpha\beta} \dot{\epsilon}_{\gamma\gamma} \right) + \tau_{\alpha\gamma} \dot{R}_{\beta\gamma} + \tau_{\gamma\beta} \dot{R}_{\alpha\gamma} \quad (13)$$

where  $G$  denotes the shear modulus,  $\dot{\epsilon}_{\alpha\beta}$  represents the strain rate tensor, and  $\dot{R}_{\alpha\beta}$  is the rotation rate tensor. The strain rate tensor and the rotation rate tensor, along with their discretized SPH formulations, are defined as follows:

$$\dot{\epsilon}_{\alpha\beta} = \frac{1}{2} \left( \frac{\partial v_\alpha}{\partial x_\beta} + \frac{\partial v_\beta}{\partial x_\alpha} \right) \quad (14)$$

$$\dot{R}_{\alpha\beta} = \frac{1}{2} \left( \frac{\partial v_\alpha}{\partial x_\beta} - \frac{\partial v_\beta}{\partial x_\alpha} \right) \quad (15)$$

In SPH form, these tensors are given by:

$$\dot{\epsilon}_{\alpha\beta i} = \frac{1}{2} \sum_j \frac{m_j}{\rho_j} \left[ v_{\alpha ji} \frac{\partial W_{ij}(h_i)}{\partial x_{\beta i}} + v_{\beta ji} \frac{\partial W_{ij}(h_i)}{\partial x_{\alpha i}} \right] \quad (16)$$

$$\dot{R}_{\alpha\beta i} = \frac{1}{2} \sum_j \frac{m_j}{\rho_j} \left[ v_{\alpha ji} \frac{\partial W_{ij}(h_i)}{\partial x_{\beta i}} - v_{\beta ji} \frac{\partial W_{ij}(h_i)}{\partial x_{\alpha i}} \right] \quad (17)$$

The second invariant of the shear stress,  $J_2$  is defined as

$$J_2 = \frac{1}{2} \tau_{\alpha\beta} : \tau_{\alpha\beta} \quad (18)$$

The von Mises stress,  $\sigma_{vm}$ , is calculated as  $\sqrt{3J_2}$ . The plastic state is determined by comparing the von Mises stress  $\sigma_{vm}$  with the yield stress  $\sigma_Y$ . When  $\sigma_{vm}$  exceeds  $\sigma_Y$ , the material particle is deemed to be in the plastic state, and the shear stress is adjusted back to the yield surface.

$$\tau_{\alpha\beta} = \begin{cases} \frac{\sigma_Y}{\sigma_{vm}} \tau_{\alpha\beta}, & \sigma_{vm} > \sigma_Y; \\ \tau_{\alpha\beta}, & \sigma_{vm} \leq \sigma_Y. \end{cases} \quad (19)$$

The Johnson–Cook model is selected for determining the yield stress  $\sigma_Y$ . This model, represented by (20), accounts for strain hardening, high strain rates, and the thermal softening characteristic of the material (Johnson 1983).

$$\sigma_Y = (A + B(\varepsilon_p)^n) \left( 1 + C \ln \frac{\dot{\varepsilon}_p}{\dot{\varepsilon}_0} \right) (1 - T^{*m}) \quad (20)$$

where A, B, C, n, m are constants;  $\varepsilon_p$  represents the effective plastic strain;  $\dot{\varepsilon}_p$  stands for the effective plastic strain rate, where  $\dot{\varepsilon}_0$  equals  $1s^{-1}$ ; and  $T^*$  denotes the dimensionless temperature, which is defined as follows:

$$T^* = \frac{T - T_{room}}{T_{melt} - T_{room}} \quad (21)$$

Plastic strain is updated when the particle exists in plastic phase. The increment in plastic strain is defined according to (22).

$$\Delta\varepsilon_p = \frac{\sigma_{vm} - \sigma_Y}{3G} \quad (22)$$

For concrete material, HJC yield model is selected instead (Ren et al. 2017). The HJC model was developed to account for high-velocity impact loads, like those generated by explosions, and is primarily used to describe material behavior under compressive loading.

$$\sigma^* = \begin{cases} [A(1-D) + BP^{*N}](1 + C \ln \varepsilon^*) \leq S_{max}, & P^* \geq 0 \\ \left[ \frac{AP^*}{T^*(1-D)} + A \right], & P^* < 0 \end{cases} \quad (23)$$

$\sigma^* = \sigma_Y/f_c$ .  $f_c$  is the compressive strength of concrete,  $S_{max}$  is the normalized maximum strength that the concrete material can withstand.  $P^* = P/f_c$  denotes the normalized pressure. A, B, N, C are constant. D is the accumulated damage determined by

$$D = \sum \frac{\Delta\varepsilon_p + \Delta\mu_p}{\varepsilon_p^f + \mu_p^f} \quad (24)$$

where  $\Delta\varepsilon_p$  and  $\Delta\mu_p$  are the effective plastic strain increment and plastic volumetric strain during a cycle of integration, respectively.  $\varepsilon_p^f + \mu_p^f$  is the total plastic strain under a constant pressure until fracture, which is determined by follows (25).

$$\varepsilon_p^f + \mu_p^f = D_1(P^* + T^*)^{D_2} \geq \text{EFMIN} \quad (25)$$

where  $T^* = T/f_c$  is the normalized tensile strength and T is the tensile strength.  $D_1$ ,  $D_2$ , EFMIN are constant.

## 2.4 The EOS of materials

The EOS is a fundamental equation that provides a relationship between thermodynamic quantities for a given system and helps predict the changes in material properties, such as density and internal energy that are critical to understanding detonation and impact dynamics. It can also be used to accurately model the behavior of materials under the extreme loading conditions that occur during impact and penetration events. Various forms of EOS are used depending on the specific requirements of the study.

The Jones–Wilkins–Lee (JWL) EOS is employed to characterize the pressure resulting from the detonation (Pramanik and Deb 2015):

$$P = A \left( 1 - \frac{\omega\eta}{R_1} \right) e^{-\frac{R_1}{\eta}} + B \left( 1 - \frac{\omega\eta}{R_2} \right) e^{-\frac{R_2}{\eta}} + \omega\eta\rho_0 E \quad (26)$$

where A, B,  $R_1$ ,  $R_2$ , and  $\omega$  represent constant coefficients dependent on the material;  $\eta$  is the ratio of the explosive material's density to the initial density;  $\rho_0$  denotes the initial density of the explosive material; and E refers to the specific internal energy.

The Mie–Gruneisen EOS is employed to characterize the pressure pertaining to solid metal materials:

$$P = \frac{\rho c_0^2(\eta - 1) \left[ \eta - \frac{\Gamma}{2}(\eta - 1) \right]}{[\eta - S(\eta - 1)]^2} + \Gamma E \quad (27)$$

Here,  $\rho$  represents the density of the material;  $c_0$  stands for the speed of sound in the material;  $\eta$  is the ratio of the density to the initial density; E denotes the specific internal energy; and  $\Gamma$ , S are the constant coefficients dependent on material.

The Holmquist Johnson Cook (HJC) EOS is utilized to depict the pressure-compaction response for the concrete material. This equation consists of three distinct phases, illustrated in Fig. 4. The initial phase, referred to as OA, follows a linear elastic response from the negative pressure cutoff  $-T(1-D)$  to the elastic limit  $P_{crush}$  and is articulated as follows:

$$P = K\mu; P < P_{crush} \quad (28)$$

where the standard volumetric strain is  $\mu = \rho/\rho_0 - 1$ ,  $K = P_{crush}/\mu_{crush}$  is the elastic bulk modulus, and  $\mu_{crush}$  is the volumetric strain at the elastic limit  $P_{crush}$ .

The subsequent phase, termed as AB, corresponds to the transitional region. During this phase, air voids are progressively compressed out of the concrete, while plastic volumetric strain induces compaction damage, continuing until the point  $(\mu_{plock}, P_{lock})$ . It is expressed as follows:

$$P = \begin{cases} P_{crush} + K_{lock}(\mu - \mu_{crush}) & \text{loading;} \\ P_0 - [(1-F)K + FK_1](\mu_0 - \mu) & \text{unloading;} \end{cases} \quad (29)$$

where  $K_{lock} = (P_{lock} - P_{crush})/(\mu_{plock} - \mu_{crush})$  is the slope, and  $\mu_{plock}$  is the volumetric strain at the compacting pressure  $P_{lock}$ .  $F = (\mu_0 - \mu_{crush})/(\mu_{plock} - \mu_{crush})$  is the interpolation factor.  $P_0$  is the pressure when the volumetric strain reached  $\mu_0$  prior to

unloading. The final phase, known as BC, represents the compact region where all air voids are considered to be extricated from the concrete. In this phase, the material is assumed to exhibit completely non-linear elasticity, and this relationship is portrayed by the ensuing equation:

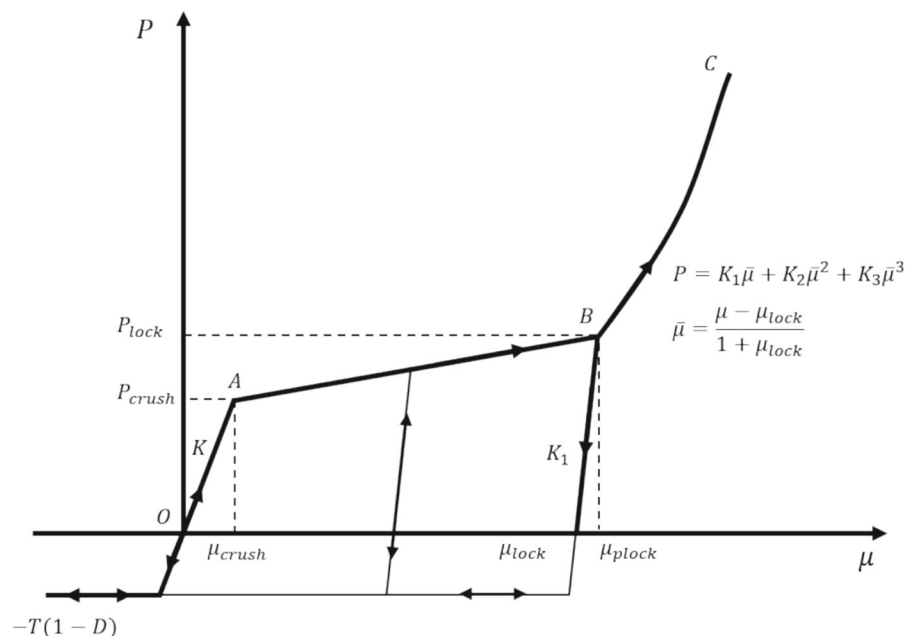
$$P = \begin{cases} K_1\bar{\mu} + K_2\bar{\mu}^2 + K_3\bar{\mu}^3 & \text{loading;} \\ K_1\bar{\mu} & \text{unloading;} \end{cases} \quad (30)$$

where  $\bar{\mu} = (\mu - \mu_{lock})/(1 + \mu_{lock})$  is the modified volumetric strain, and  $\mu_{lock} = \rho_{grain}/\rho_0 - 1$  is the locking volumetric strain.  $\rho_{grain}$  is the material's grain density, absent of any air voids.  $K_1$ ,  $K_2$ , and  $K_3$  are the material constants.

Determination of loading/unloading conditions was achieved by evaluating the plastic multiplier  $\lambda$  within the computation.  $\lambda = 0$  signifies continued loading, whether the stress point still lies inside the elastic domain or has already reached the plastic yield surface.  $\lambda > 0$  denotes unloading while the material is in a plastic state. In the HJC model, 'lambda' is defined as follows

$$\lambda = \frac{-K\dot{\epsilon}^{\gamma\gamma}\left(\frac{B}{N}\frac{P^{N-1}}{f_c^N}\right) + \frac{G}{\sqrt{J_2}}\frac{\sqrt{3}}{f_c}\tau^{a\beta}\dot{\epsilon}^{a\beta}}{K\left(\frac{B^2}{N^2}\frac{P^{2N-2}}{f_c^{2N}}\right) + G\frac{3}{f_c^2}} \quad (31)$$

**Fig. 4** Holmquist Johnson Cook equation of State (Jin et al. 2017)



## 2.5 Computing algorithm

Our GPU-SPH solver is constructed based on an open-source, multi-physics SPH code known as SOPHIA (Jo et al. 2019, 2022; Park et al. 2020), which is developed in CUDA C language. Figure 5 illustrates the algorithm utilized in our study, which is based on CPU-GPU heterogeneous computing. For initialization and output saving, the CPU is utilized, while the GPU accelerates the computation of governing equations. After initializing the simulation, memory is transferred from the CPU to the GPU. Particles are then sorted based on uniform rectangular cells that are larger than each particle's smoothing length. The simulation progresses through computing steps where particles solve discretized governing equations.

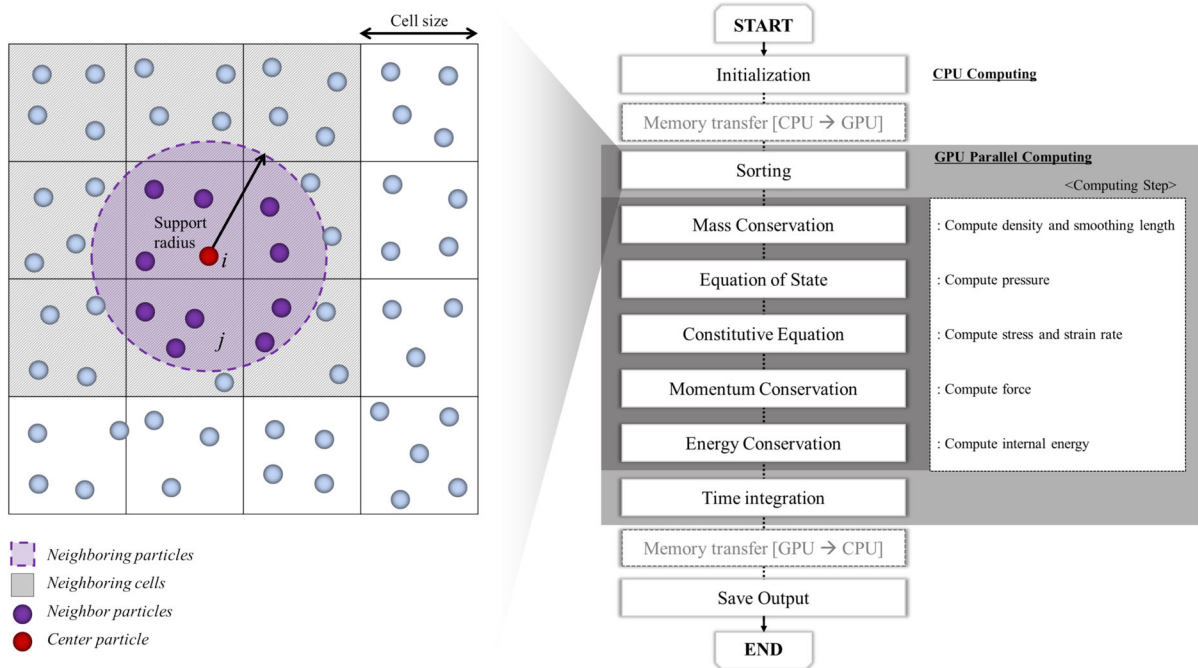
In this setup, each thread is assigned to one particle, allowing for concurrent computations. Each thread checks surrounding particles in adjacent cells to determine if they are neighbors and then performs calculations accordingly as in Fig. 5 (Jo et al. 2019). The density is calculated based on the mass of neighboring particles. For each particle, the smoothing length is adjusted in response to density variations. These adjustments occurring through several

iterations within each thread, which is allocated to an individual particle. Pressure is calculated using the EOS, which depends on density and internal energy. Additionally, the constitutive equation is solved to compute the shear stress tensor and strain rate. Subsequently, the momentum and energy equations are solved to update the acceleration and internal energy of the particles. The governing equations are integrated over time, and in this study, we adopted a predictor–corrector time integration scheme.

This process is repeated for each time step until the simulation reaches the predetermined end time. After the computation concludes or at periodic intervals, data is transferred back to the CPU memory, and output files are generated.

## 2.6 GPU implementation

GPU acceleration was introduced because SPH demands extremely small time-steps and high particle resolutions, making serial CPU execution prohibitive in both space and time. In the implementation, every SPH particle is mapped to a single CUDA thread, which is executed within a streaming multiprocessor (SM) composed of multiple CUDA cores. This



**Fig. 5** The overall algorithm of the GPU-SPH solver

mapping strategy and the hierarchical CUDA architecture are illustrated in Fig. 6. This parallel mapping structure allows all governing equations to be computed in parallel according to the GPU architecture.

The overall simulation process is divided into three stages: initialization, solution, and output (Jo et al. 2022). The initialization step such as reading input and setting parameters is performed on the CPU (host). Particle data are then transferred to the GPU (device), where the main numerical steps including mass and momentum conservation and contact force estimation are solved in parallel for each particle. After time integration, the resulting data are copied back to the CPU for post-processing and output generation.

To sustain high throughput, a uniform-grid sorting scheme reorders particles by cell index so that neighbors reside contiguously in memory, boosting cache-hit rates during SPH neighbor search.

This sorting algorithm proceeds in three steps (Jo et al. 2022): (1) computing a grid cell index for each particle, (2) reordering the particle array based on the

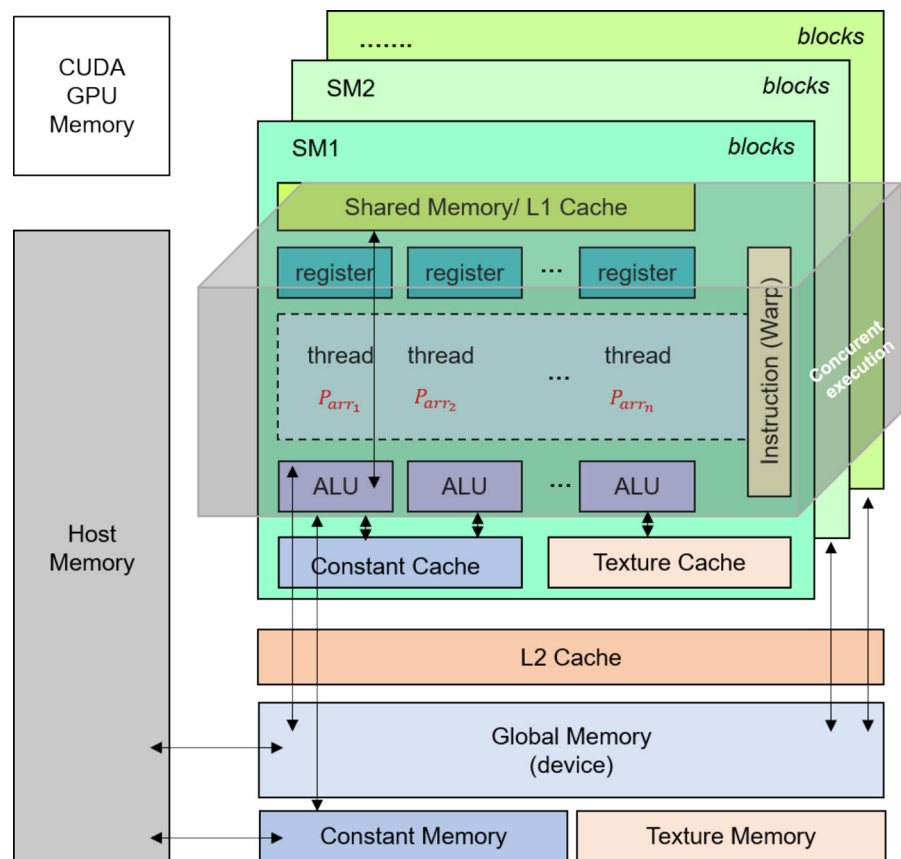
ascending grid index, and (3) constructing cell start and end index arrays. This enables efficient identification of neighboring particles by limiting the neighbor search to adjacent cells. As a result, memory locality is significantly improved, which increases cache efficiency and reduces memory latency during SPH neighbor search.

All simulations were executed on an NVIDIA GeForce GTX 1080 Ti, thereby making large-scale SPH studies computationally feasible.

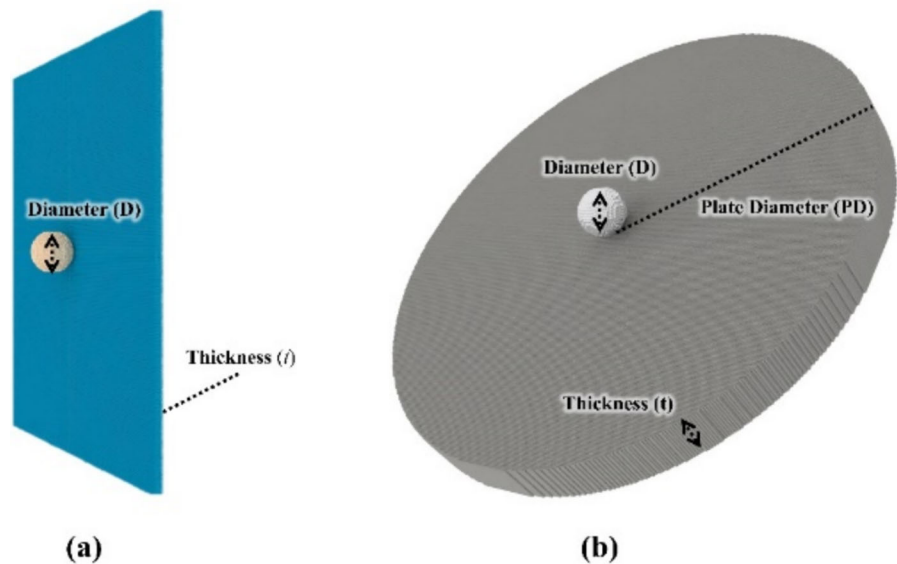
### 3 Numerical validation

In this study, validation was carried out in two stages to demonstrate the e of our method in handling intense collision phenomena between ductile metal projectiles and brittle concrete materials. Initially, to establish our method's capability in typical collision scenarios, numerical simulations on high-velocity impact (HVI) analyses were performed on ductile targets such as

**Fig. 6** CUDA GPU memory structure



**Fig. 7** Initial configuration of numerical simulations for: **a** metal–metal HVI, and **b** metal–concrete HVI



aluminum and copper. This was followed by further validation through simulations of HVIs on brittle concrete. These steps successfully confirmed the versatility of our methodology in addressing a spectrum of materials from ductile metals to brittle concrete.

### 3.1 HVI on metal material

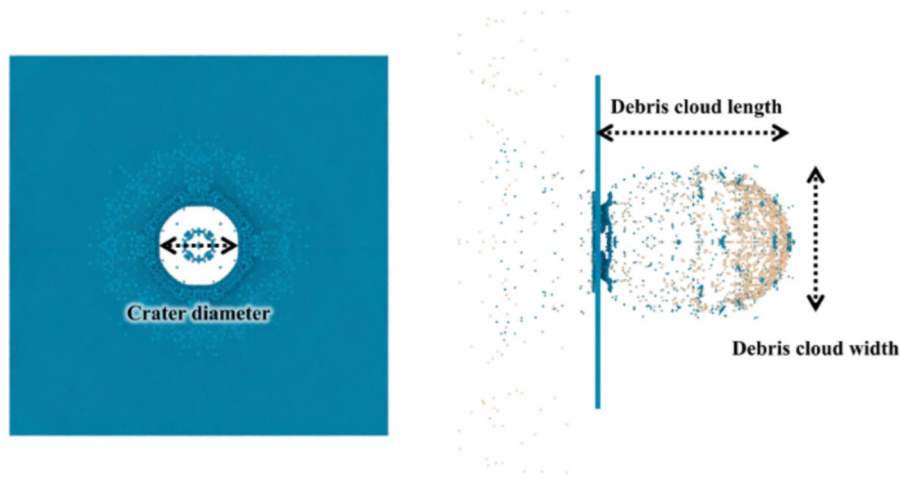
In this section, we examine two scenarios of high-velocity impacts (HVI) where an aluminum ball, 1 cm in diameter ( $D$ ), was launched at a thin plate. In the first scenario, the ball struck an aluminum plate with a thickness ( $t$ ) of 0.4 cm at a velocity of 6.18 km/s. In the second scenario, it impacted a copper plate with a thickness ( $t$ ) of 0.15 cm at a velocity of 5.75 km/s. The simulations deployed approximately 450,000 SPH particles for the aluminum target and 120,000 for the copper target. The initial spacing between particles was set at 0.5 mm; the initial smoothing length was 1.2 times the particle spacing; and the time step interval was configured to 0.01 microseconds, and each simulation ran up to an end time of 20 microseconds. The GPU-parallelized computations efficiently completed each scenario, taking approximately 20 min for the aluminum target with 450,000 SPH particles and about 10 min for the copper target with 120,000 SPH particles. In contrast, the same simulations using a CPU-only setup required approximately 8 and 6 h, respectively, two simulation scenarios were

**Table 2** Parameter of the Mie-Gruneisen model, Johnson–Cook model and property for Aluminum (Chen et al. 2019) and Copper (Riabinkina et al. 2022)

| Parameter                                     | Aluminum | Copper |
|-----------------------------------------------|----------|--------|
| Initial density $\rho_0$ (kg/m <sup>3</sup> ) | 2710     | 8900   |
| Speed of sound $c_0$ (m/s)                    | 5300     | 3940   |
| Shear modulus $G$ (GPa)                       | 27.6     | 44.7   |
| Mie-Gruneisen coefficient $S$                 | 1.5      | 1.489  |
| Mie-Gruneisen coefficient $\Gamma$            | 1.7      | 2.02   |
| Johnson–Cook coefficient $A$ (MPa)            | 175      | 90     |
| Johnson–Cook coefficient $B$ (MPa)            | 380      | 292    |
| Johnson–Cook coefficient $C$                  | 0.0015   | 0.025  |
| Johnson–Cook coefficient $n$                  | 0.34     | 0.31   |
| Johnson–Cook coefficient $m$                  | 1.0      | 1.09   |
| Reference temperature $T_{room}$ (K)          | 273      | 293    |
| Melting point $T_{melt}$ (K)                  | 775      | 1356   |

conducted, with the geometry details provided in Fig. 7a. The Mie-Gruneisen equation of state (EOS) model was adopted for computing pressure, and the Johnson–Cook constitutive model was utilized for material behavior with the parameters summarized in Table 2. Simulation results were compared with experimental outcomes obtained under identical conditions and previous simulation findings.

In these simulations, key metrics such as crater diameter and the debris cloud aspect ratio—defined as



**Fig. 8** Representative parameter from HVI simulations

**Table 3** Comparison between experiment and simulation results of HVI on an aluminum plate

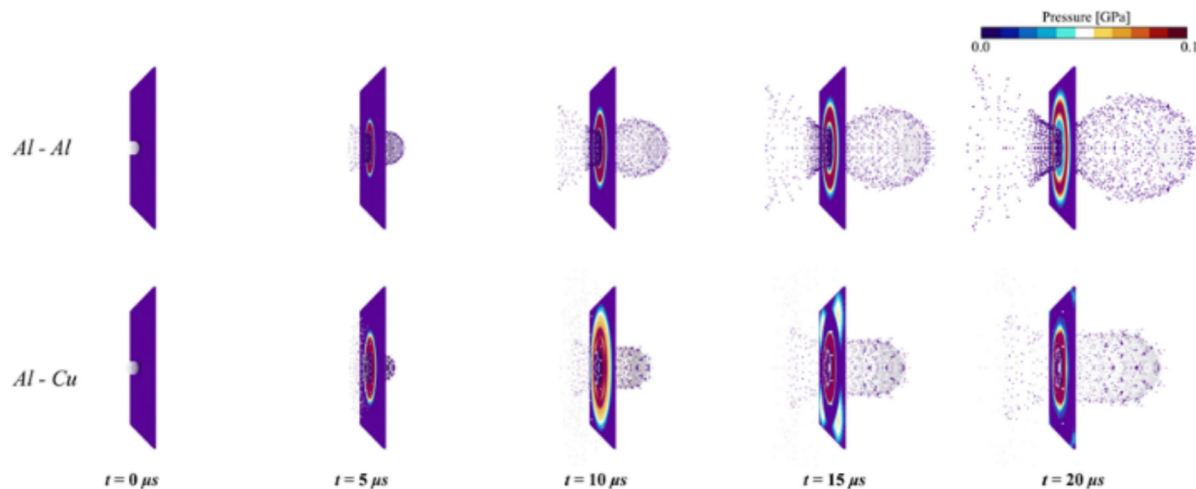
|                                          | Crater diameter (cm) | Debris cloud aspect ratio     | Error        |
|------------------------------------------|----------------------|-------------------------------|--------------|
| Experiment (Hiermaier et al. 1997)       | 3.1 (2.75 ~ 3.45)    | 1.39                          |              |
| Simulation (SPH) (Hiermaier et al. 1997) | 3.5                  | 1.11                          | 12.9%, 20.1% |
| Simulation (SPH) (Mehra et al. 2006)     | 2.8                  | 1.53                          | 9.7%, 10.1%  |
| Simulation (SPH) (Xu et al. 2025)        | 2.69                 | 1.258                         | 13.2%, 9.5%  |
| This study (SPH)                         | 2.8                  | 1.328 ( $\frac{9.3cm}{7cm}$ ) | 9.7%, 4.5%   |

**Table 4** Comparison between experiment and simulation results of HVI on a copper plate

|                                          | Crater diameter (cm) | Debris cloud aspect ratio        | Error        |
|------------------------------------------|----------------------|----------------------------------|--------------|
| Experiment (Hiermaier et al. 1997)       | 2.12                 | 1.39                             |              |
| Simulation (SPH) (Hiermaier et al. 1997) | 2.3                  | 1.33                             | 8.5%, 4.3%   |
| Simulation (SPH) (Mehra et al. 2006)     | 2.3                  | 1.7                              | 8.5%, 22.3%  |
| Simulation (SPH) (Lu et al. 2023)        | 2.27                 | 1.37                             | 6.6%, 1.4%   |
| Simulation (SPH) (Xu et al. 2025)        | 2.03                 | 1.3739                           | 4.3%, 1.2%   |
| This study (SPH)                         | 2.1                  | 1.389 ( $\frac{7.92cm}{5.7cm}$ ) | 0.94%, 0.07% |

the length of the debris cloud divided by its width—were evaluated as defined in Fig. 8. An aspect ratio value of 1 signifies a perfectly circular shape, with values greater than 1 reflecting more elongated debris clouds. Given the constant collision energy—defined by the mass and velocity of the impacting entity across both simulations and experiments—the observed lower aspect ratios in simulations imply a diminished conversion of collision energy into thermal energy,

relative to actual events. Our results exhibited less error with the experiment than the previous simulation results (Hiermaier et al. 1997) as shown in Tables 3 and 4. The results from other studies in Table 3 show crater diameters similar to ours but differ in debris cloud aspect ratio, whereas those in Table 4 show similar debris cloud shapes but differ in crater diameter. This pattern reflects the material configurations: for the aluminum plate (same-material impact),



**Fig. 9** Dynamic response of the debris cloud from the target and projectile in a HVI simulation

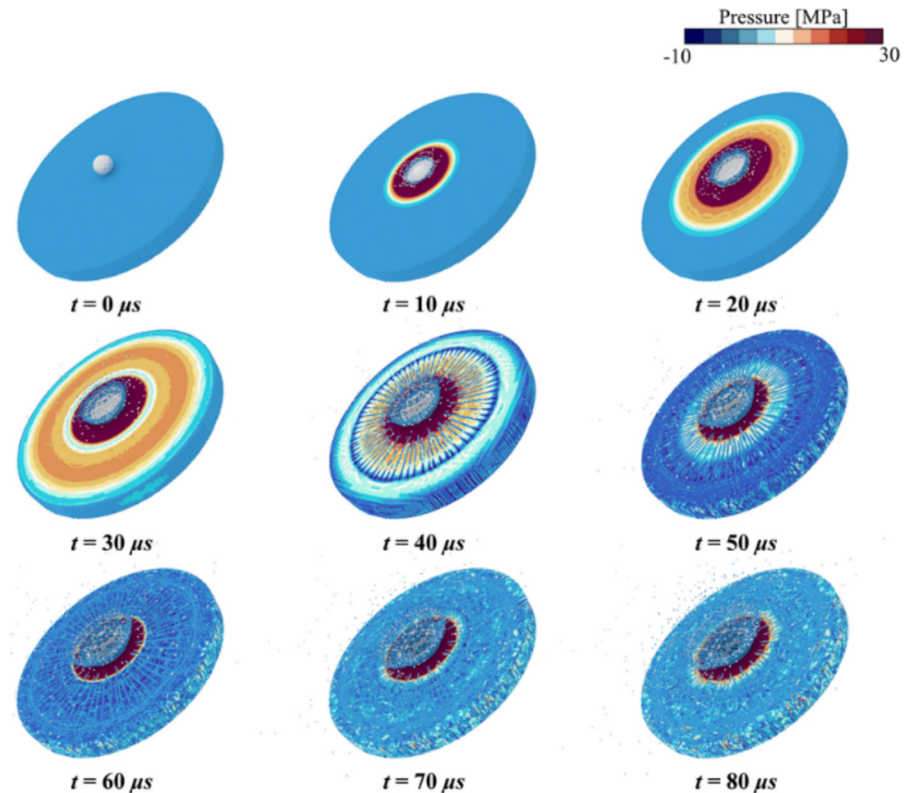
**Table 5** Parameter of the Holmquist Johnson Cook (HJC) model for 150 MPa concrete material (Chen et al. 2023)

| Parameter                                      | Concrete            |
|------------------------------------------------|---------------------|
| Compressive strength $f_c$ (MPa)               | 150                 |
| Initial density $\rho_0$ (kg/m <sup>3</sup> )  | 2700                |
| Shear modulus $G$ (GPa)                        | 20.16               |
| Normalized cohesive strength $A$               | 0.79                |
| Normalized pressure hardening $B$              | 1.6                 |
| Strain rate coefficient $C$                    | 0.007               |
| Pressure hardening exponent $N$                | 0.61                |
| Maximum tensile hydrostatic pressure $T$ (MPa) | 7.59                |
| Normalized maximum strength $S_{\max}$         | 7                   |
| Crushing pressure $P_{\text{crush}}$ (MPa)     | 16                  |
| Crushing Volumetric strain $U_{\text{crush}}$  | 0.001               |
| Locking pressure $P_{\text{lock}}$ (MPa)       | $8 \times 10^2$     |
| Locking volumetric strain $U_{\text{lock}}$    | 0.1                 |
| Damage constant $D_1$                          | 0.04                |
| Damage constant $D_2$                          | 1                   |
| Pressure constant $K_1$ (MPa)                  | $8.5 \times 10^4$   |
| Pressure constant $K_2$ (MPa)                  | $-17.1 \times 10^4$ |
| Pressure constant $K_3$ (MPa)                  | $20.8 \times 10^4$  |

the smoothing length adjustment plays a key role, while for the copper plate (multi-material impact), the unified material model becomes more influential. Notably, the simulation of the copper plate collision closely matched the experimental data, signifying a minimal error, and allowed for the validation of the material model for copper, which is the material used in the shaped charge jet to be analyzed in the next section. The accuracy of the present method stems

from retaining the material parameters given by (Hiermaier et al. 1997), selecting a particle spacing that can resolve both pre- and post-shock density gradients. A support domain large enough to give each particle sufficient neighbors, together with the Wendland C2 kernel, suppresses pairing instability and prevents numerical divergence under the extreme pressure and temperature gradients generated by the impact (Fletcher et al. 2015). Figure 9 illustrates the

**Fig. 10** Pressure distribution of the HVI on concrete at different time instants with a 10 microsecond interval



time evolution of the debris cloud for both scenarios. The simulations conducted on plates made of different materials—aluminum and copper—exhibit distinct dynamic behaviors. The more spherical debris clouds observed with aluminum may be attributed to its lower density, shear modulus, and melting point compared to copper.

### 3.2 HVI on concrete material

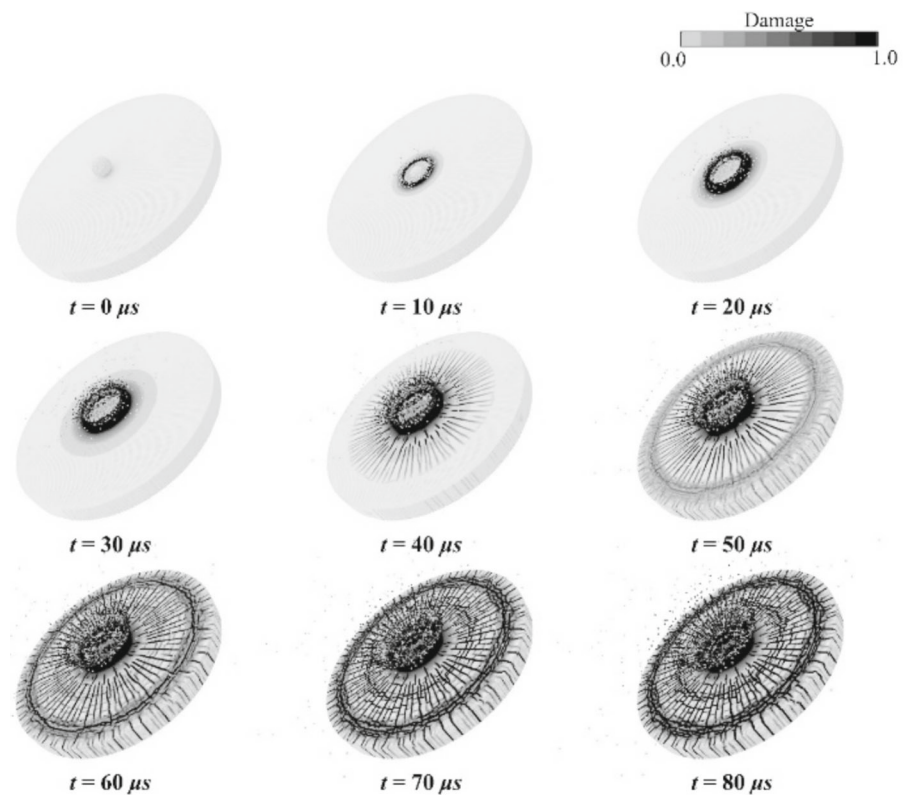
Compared to a ductile material like metal, a brittle material such as concrete is more likely to suffer severe damage due to its tendency to fracture and break under stress, while ductile materials can deform and absorb more energy through plastic deformation before failing. Upon impact, brittle materials experience stress concentration at the point of contact, resulting in rapid crack propagation and failure with minimal deformation. In contrast, ductile materials are capable of extensive plastic deformation, which allows them to absorb significant amounts of energy and thus exhibit greater resistance to damage.

Similar to the previous metal HVI simulation, a new scenario was carried out under conditions

identical to those in the simulation by (Hiermaier et al. 1997) and the corresponding experiment, with results compared for validation. In this simulation, an aluminum ball with a diameter of 2.54 cm was launched at a speed of 2 km/s towards a concrete cylinder-shaped plate with a compressive strength ( $f_c$ ) of 150 MPa. The concrete plate had a diameter of 30 cm and a thickness of 2.4 cm. The parameters of the HJC model used for this simulation are detailed in Table 5. The simulation utilized approximately 1,700,000 SPH particles. The initial particle spacing was set at 1 mm, and the initial smoothing length was 1.5 times the initial particle spacing. The time step interval was 100 micro seconds, with a simulation end time of 100 micro seconds. Utilizing GPU acceleration, our simulation completed in just 1 h, significantly faster than the 2.5 h reported in previous simulation (Chocron et al. 2019) under the same conditions and the 35 h required when using a CPU-only setup.

Figures 10 and 11 illustrate the numerical predictions of the time evolution of the dynamic behaviors of a concrete plate and an aluminum projectile during a high-velocity impact (HVI). The figures capture the

**Fig. 11** Damage distribution of the HVI on concrete at different time instants with a 10 microsecond interval



**Table 6** Comparison between experiment and simulation results of HVI on a concrete plate

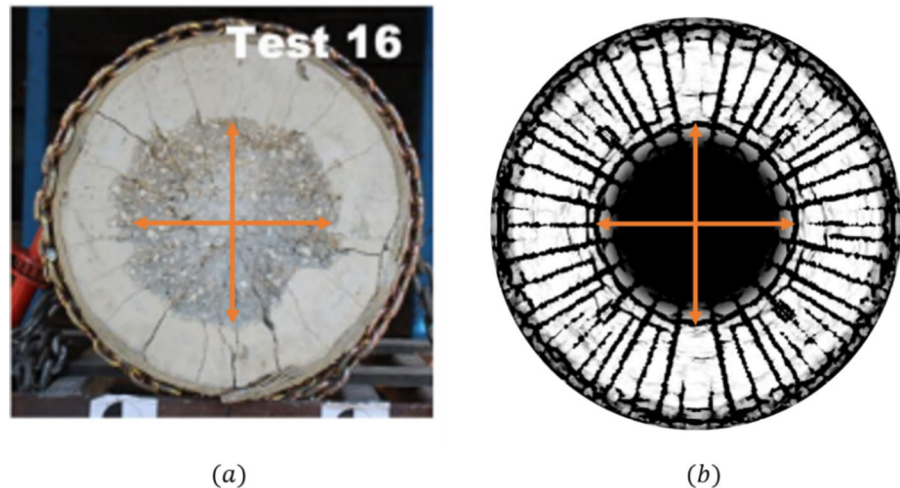
|                                     | Crater diameter (mm) | Error  |
|-------------------------------------|----------------------|--------|
| Experiment (Chocron et al. 2019)    | 80                   |        |
| Simulation (SPH) (Chen et al. 2023) | 85                   | 6.25%  |
| This study (SPH)                    | 80                   | < 0.1% |

scene at 10-microsecond intervals, with pressure and damage visually represented. As depicted in Fig. 10, the impact between the aluminum ball and the concrete plate generates a shockwave that propagates from the center to the edges of the plate. After the shockwave reflects, the pressure begins to decrease from the edges toward the center. (see supplementary\_video1) The shockwave and subsequent stress wave lead to the development and spread of cracks and fractures in the concrete. Initially, 20 radial cracks form and gradually spread, as shown in Fig. 11. This stress wave reflects back, forming a circular crack

visible from 50 microseconds onward. By 80–90 microseconds, this circular crack has branched out, and radial cracks have reached the edges of the concrete plate. (see supplementary\_video2).

The diameter of the completely damaged area was compared against the findings presented by (Chen et al. 2023), as summarized in Table 6 and Fig. 12. The results confirm that this study successfully replicated the experimental outcomes, which were obtained from an experiment originally reported by (Chocron et al. 2019) that involved hypervelocity impacts ( $\sim 2$  km/s) of aluminum projectiles onto high strength concrete to investigate momentum enhancement under extreme loading conditions. A critical factor in the accuracy of this study was the inclusion of an artificial viscosity term in the energy conservation (10). This term was carefully designed to prevent over-compression in the rate of energy change, allowing the simulation to accurately handle the large deformations of both metal and concrete materials while closely adhering to the principle of energy conservation. This validation underscores the effectiveness of our SPH method in managing the

**Fig. 12** Comparison of crater diameter between **a** experimental observation ((Chocron et al. 2019) and **b** SPH simulation



**Table 7** Parameter of JWL EOS model for PTX8701 (Hu et al. 2017)

| Parameter                                                 | PTX8701              |
|-----------------------------------------------------------|----------------------|
| Reference density $\rho_0$ (kg/m <sup>3</sup> )           | 1710                 |
| Detonation velocity $D$ (m/s)                             | 7980                 |
| Chapman-Jouguet pressure $P_{CJ}$ (GPa)                   | $2.86 \times 10^4$   |
| Constant $A$ (MPa)                                        | $5.2423 \times 10^5$ |
| Constant $B$ (MPa)                                        | $7.768 \times 10^3$  |
| Constant $R_1$                                            | 4.2                  |
| Constant $R_2$                                            | 1.1                  |
| Constant $w$                                              | 0.34                 |
| Detonation energy per unit volume $E$ (J/m <sup>3</sup> ) | $8.5 \times 10^9$    |

significant deformation challenges presented by metal and concrete materials.

#### 4 Shaped charge (SC) jet penetration into concrete cylinder

In this section, as a final benchmark, the collision of a shaped charge jet into a concrete cylinder is simulated. The simulation results are compared with experimental observations based on representative length parameters. Additionally, variables such as compressive strength and shaped charge geometry are altered to demonstrate the capability of our SPH method to produce reasonable results under various conditions.

#### 4.1 Numerical simulation setup

A Shaped charge (SC) is constructed from a cylindrical explosive that has a conical cavity, flanked by a thin metal liner on one side and a booster on the opposite side. The detonation of the explosive by the detonator initiates an outward spherical detonation waves. This action concurrently propels the metallic liner material due to the intense detonation pressure, leading to the formation of a high-velocity liner jet. As a result, the target undergoes damage from the exceedingly high kinetic energy produced by the high-velocity jet. This occurrence represents a type of fluid–structure interaction (FSI) analysis and requires high accuracy, as the high-velocity jet acts as fluid and interacts with the structure.

In this study, SPH simulations were conducted using a benchmark problem, for which experimental data and FEM simulation data from (Hu et al. 2017) are available. The explosive material, PTX8701, was modeled using the JWL equation of state (EOS), with parameters summarized in Table 7. Additionally, the properties of the copper liner are detailed in Table 2 from the previous section, and the properties of the concrete cylinder are outlined in Table 8. As illustrated in Fig. 13, the shaped charge (SC) featured a cone-shaped copper liner with a 120-degree angle and a thickness of 0.135 cm. The SC's diameter and length were both set at 5.6 cm. The initial particle spacing for the SC was established at 0.05 cm, because results with larger spacing were observed to yield poor jet formation.

**Table 8** Parameter of the Holmquist Johnson Cook (HJC) model for concrete material (Hu et al. 2017)

| Parameter                                      | Concrete             |                      |                      |
|------------------------------------------------|----------------------|----------------------|----------------------|
| Compressive strength $f_c$ (MPa)               | 20                   | 44                   | 200                  |
| Initial density $\rho_0$ (kg/m <sup>3</sup> )  | 2400                 | 2400                 | 2500                 |
| Shear modulus $G$ (GPa)                        | 12.5                 | 14.86                | 24.3                 |
| Normalized cohesive strength $A$               | 0.79                 | 0.79                 | 0.3                  |
| Normalized pressure hardening $B$              | 1.6                  | 1.6                  | 1.73                 |
| Strain rate coefficient $C$                    | 0.007                | 0.007                | 0.007                |
| Pressure hardening exponent $N$                | 0.61                 | 0.61                 | 0.79                 |
| Maximum tensile hydrostatic pressure $T$ (MPa) | 2.2                  | 4                    | 20                   |
| Normalized maximum strength $S_{\max}$         | 7                    | 7                    | 7                    |
| Crushing pressure $P_{\text{crush}}$ (MPa)     | 6.9                  | 14.7                 | 66.7                 |
| Crushing Volumetric strain $U_{\text{crush}}$  | $4.1 \times 10^{-4}$ | $7.4 \times 10^{-4}$ | $2.1 \times 10^{-3}$ |
| Locking pressure $P_{\text{lock}}$ (MPa)       | $8 \times 10^2$      | $8 \times 10^2$      | $8 \times 10^2$      |
| Locking volumetric strain $U_{\text{lock}}$    | 0.1                  | 0.1                  | 0.1                  |
| Damage constant $D_1$                          | 0.04                 | 0.04                 | 0.04                 |
| Damage constant $D_2$                          | 1                    | 1                    | 1                    |
| Pressure constant $K_1$ (MPa)                  | $1.7 \times 10^4$    | $1.7 \times 10^4$    | $8.5 \times 10^4$    |
| Pressure constant $K_2$ (MPa)                  | $3.8 \times 10^4$    | $3.8 \times 10^4$    | $-1.71 \times 10^5$  |
| Pressure constant $K_3$ (MPa)                  | $2.98 \times 10^4$   | $2.98 \times 10^4$   | $2.08 \times 10^5$   |

However, for the  $f_c = 44$  MPa concrete cylinder, with a diameter of 57 cm and a height of 50 cm, conducting SPH simulations with the same initial particle spacing of 0.05 cm as used for the shaped charge would require approximately 1 billion particles. To manage computational resources more effectively, the initial particle spacing for the concrete was increased to 0.08 cm. Additionally, as illustrated in Fig. 13, the dimensions of the concrete cylinder were reduced to a diameter of 20 cm and a height of 30 cm. These adjustments allowed the simulations to be conducted with approximately 20 million particles. These modifications and their implications are further discussed in subsequent sections. The initial smoothing length was set at 1.2 times the initial particle spacing, and the time step interval was adjusted to 100 microseconds. As a result, simulating up to 300 microseconds took approximately 3 days to complete. Due to the extremely large number of particles involved, a CPU-only computation was infeasible.

#### 4.2 Formation and deformation of shape-charged jet

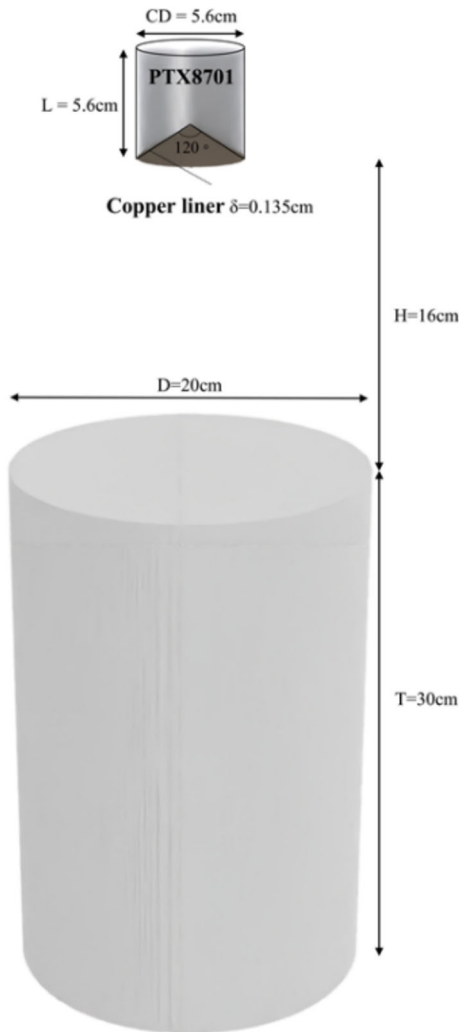
In this simulation, as shown in Fig. 14, the copper liner deformed due to detonation, forming a high-velocity

metal jet with a maximum speed of about 4,100 m/s. This jet hit the concrete and then the kinetic energy of the jet was gradually dissipated as heat and mechanical work, breaking the concrete. When the energy was sufficiently dissipated, the jet stop. For comparison with the experiment, depth of penetration (DOP), the final crater diameter  $D_c$ , and the average penetration borehole diameter  $D_h$  were measured as depicted in Fig. 15.

Although the dimensions of the concrete cylinder used in the simulations differ from those in the experimental setup, their validity is supported by rigorous analysis. The DOP is particularly crucial for evaluating the structural integrity of concrete, reflecting its impact durability, material properties, safety margins, and its role in guiding damage assessments. To ensure accurate comparisons of DOP, a sensitivity analysis was conducted on various heights and diameters. As observed in Fig. 16a, there is no significant difference in outcomes and within a 4% difference from experiment as long as the cylinder's height prevents perforation. Therefore, the height for the simulation was set at 30 cm.

Conversely, as shown in Fig. 16b, decreasing the diameter of the concrete cylinder increases the DOP. This increase occurs because a smaller diameter

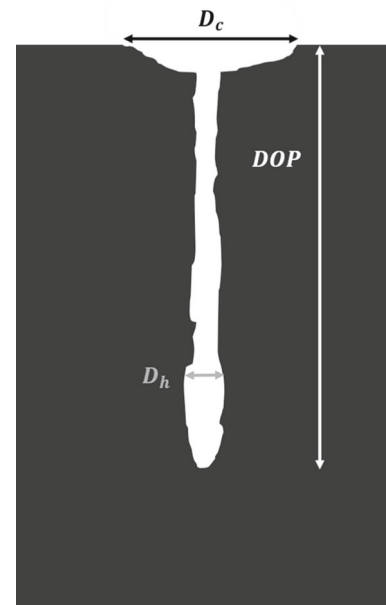
causes the shockwave and stress wave generated by the impact to reach the edges of the cylinder more rapidly, leading to quicker and more complete



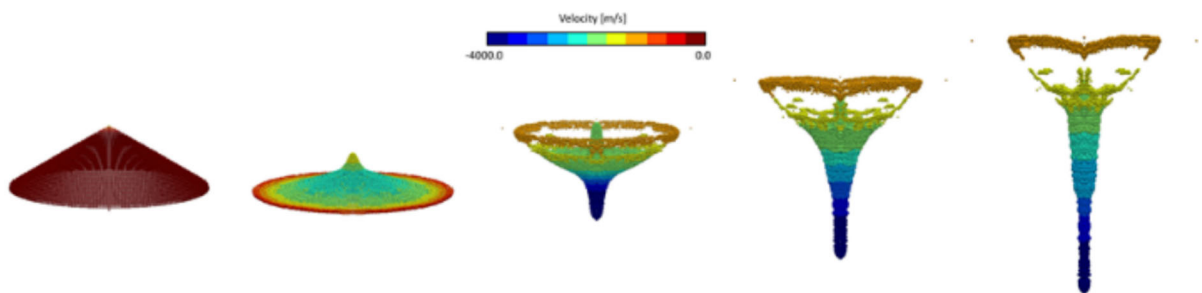
**Fig. 13** Initial configuration of shaped charge and concrete cylinder

fractures. Increasing the cylinder's diameter to converge with experimental observations showed that changes in DOP become less pronounced when the diameter exceeds 20 cm, with a difference within 4%. Therefore, the diameter for the simulations was set at 20 cm.

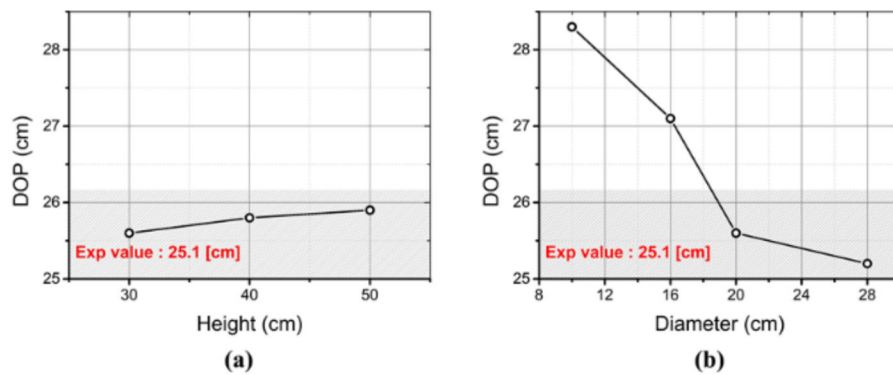
With the geometry details provided, as shown in Table 9, both the DOP and the average penetration  $D_h$  exhibited reasonable agreement with experimental results, within a 4% error margin. Additionally, the final  $D_c$  demonstrated a smaller error compared to experimental results than was observed in previous FEM simulations (Chen et al. 2023).



**Fig. 15** Representative parameters from SC jet penetration simulation



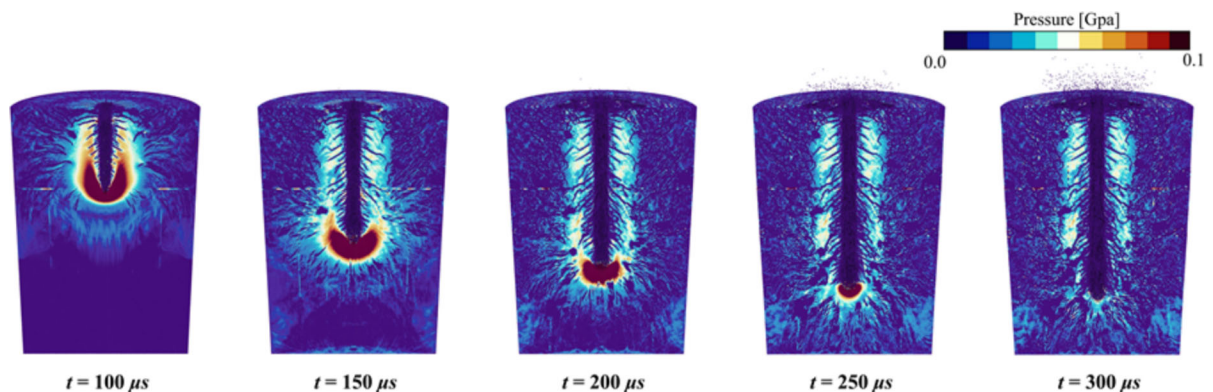
**Fig. 14** Progression of metal jet forming at different times 0, 10, 20, 30, 40  $\mu$ s



**Fig. 16** DOP of concrete cylinders with different geometry conditions: **a** concrete cylinder height and **b** concrete cylinder diameter

**Table 9** Comparison between experiment and simulation results of SC jet penetration to concrete cylinder

|                                   | $D_c$ (cm) | $D_h$ (cm) | DOP (cm) | Error             |
|-----------------------------------|------------|------------|----------|-------------------|
| Experiment (Hu et al. 2017)       | 11.2       | 3.5        | 25.1     |                   |
| Simulation (SPH) (Hu et al. 2017) | 8.0        | 3.37       | 25       | 28.6%, 3.7%, 0.4% |
| This study (SPH)                  | 9.0        | 3.36       | 25.6     | 19.6%, 4.0%, 2.0% |



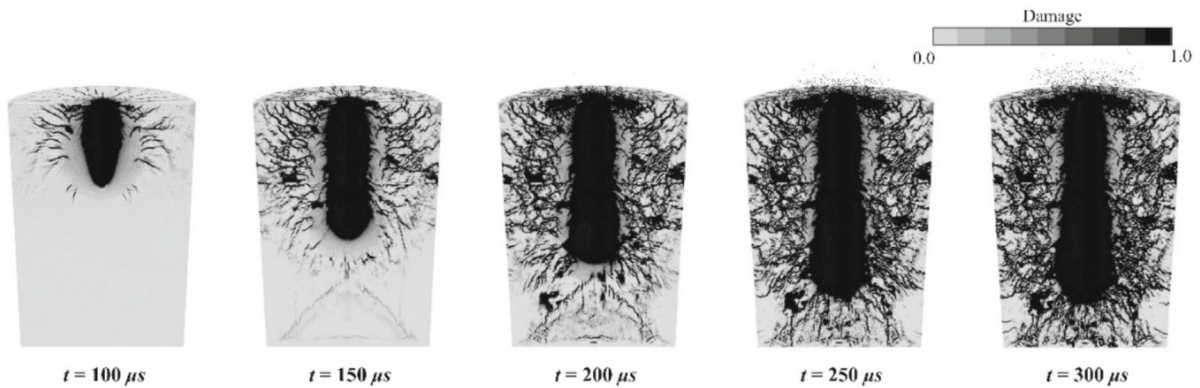
**Fig. 17** Pressure distribution of a concrete cylinder during jet penetration at different time instants: 100, 150, 200, 250, and 300  $\mu s$

#### 4.3 Damage response of the concrete cylinder

As the shaped charge jet penetration progresses, a shockwave is generated at the tip of the jet, as illustrated in Fig. 17. The range of this shockwave gradually diminishes throughout the penetration process, and eventually dissipates completely once the jet is fully consumed, as depicted in last figure in Fig. 17 (see supplementary\_video3). During the penetration process, an increase in pressure around the borehole, compared to pre-penetration levels, leads to the formation of branched cracks as displayed in Fig. 18

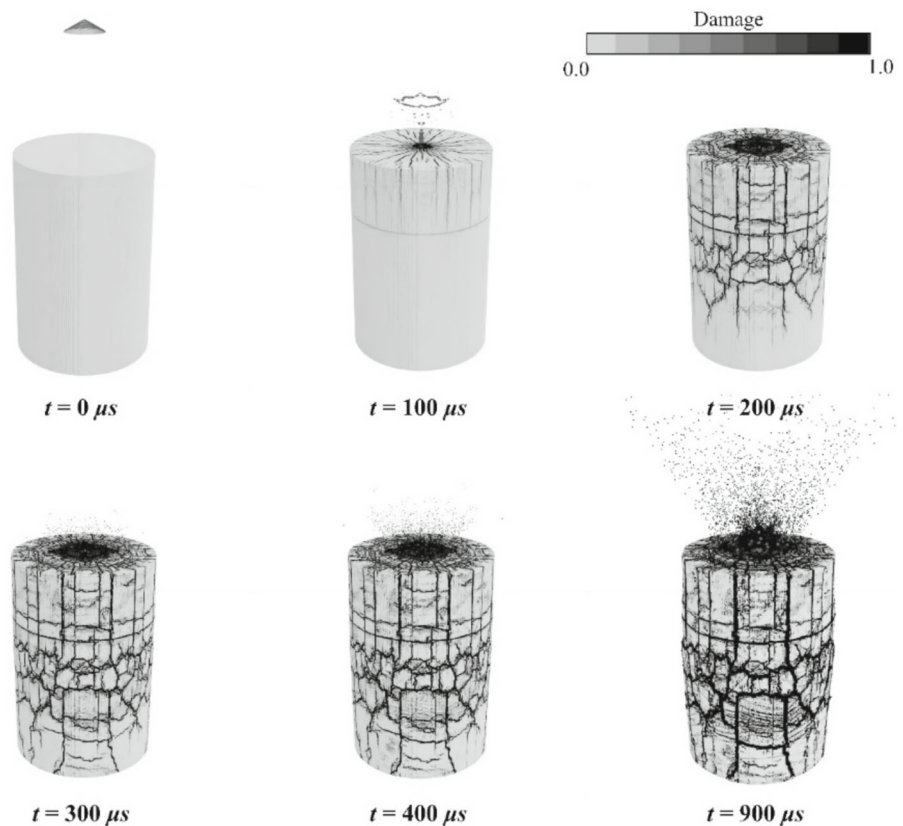
with damage highlighted (see supplementary\_video4). This indicates a complex stress distribution and material response under high-strain conditions. This interaction between the concrete and the jet underscores the structural response under extreme conditions, emphasizing the integrity of concrete structures subjected to high-velocity jet impacts.

Figure 19 illustrates the time evolution of the damage response of the concrete cylinder from a bird's eye view. It can be observed that the top face exhibited radial cracks and circular cracks due to the copper jet, indicating that the jet's strong impact



**Fig. 18** Damage distribution of a concrete cylinder during jet penetration at different time instants: 100, 150, 200, 250, and 300  $\mu\text{s}$

**Fig. 19** Damage distribution of a concrete cylinder during jet penetration at different time instants from a bird's-eye view: 0, 100, 200, 300, 400, and 900  $\mu\text{s}$

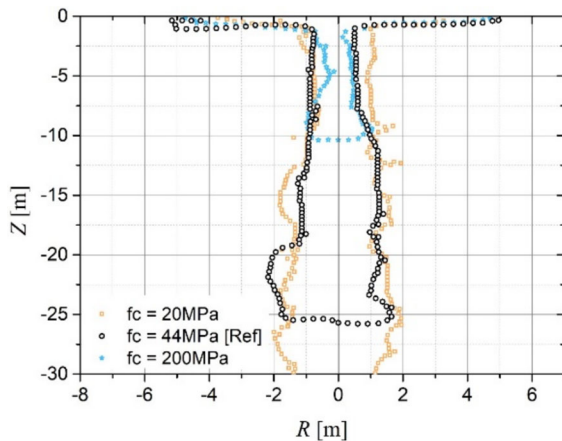


generated shockwaves and stress waves that reached and reflected off the edge of the concrete cylinder. For the cross section, cracks spread outward from the area of complete fracture near the penetration like branches. Below the penetrated area, there was a completely fractured zone extending approximately

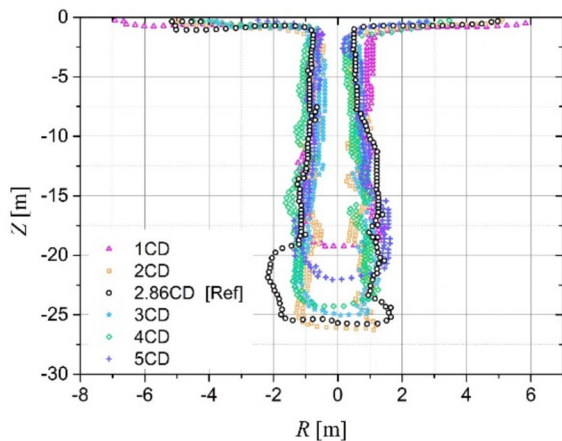
2 cm, beneath which cracks were present but did not fully penetrate to the bottom face. This demonstrates that even if the concrete structure is penetrated, it can maintain a certain level of integrity as long as it does not undergo complete perforation.

#### 4.4 Parametric analysis

Building on the verification with experimental results, this study further explored the impact of the concrete's compressive strength and the standoff distance



**Fig. 20** Borehole profiles of the concrete cylinder with different compressive strength



**Fig. 21** Borehole profiles of the concrete cylinder with different standoff distance

between the shaped charge and the concrete cylinder. As depicted in Fig. 20, when the concrete's compressive strength ( $f_c$ ) was 20 MPa, the concrete cylinder was completely penetrated, and the crater diameter was 7.08 cm. For a strength of 44 MPa, the results are presented in Table 9, and when  $f_c$  was 200 MPa, the DOP was 10.25 cm with a crater diameter of 9.36 cm. This indicates that lower concrete strength results in a larger DOP and a smaller borehole diameter. The weaker the concrete, the less resistance it offers to penetration, allowing the jet to pierce through more effectively. Consequently, the jet spends less time at the impact surface, resulting in a smaller borehole diameter. Conversely, higher concrete strength increases resistance, preventing the jet from advancing through the impact surface. This causes the kinetic energy to be transmitted radially for a longer period, thereby increasing the borehole diameter.

Simulations were conducted to examine the effects of varying the standoff distance, measured as multiples of the charge diameter (CD), as shown in Fig. 21. At a standoff distance of 1CD, the DOP was 19.13 cm, and the crater diameter was 12.77 cm. At 2CD, the DOP increased to 26.23 cm, with a crater diameter of 10.16 cm. Results for 2.86CD are detailed in Table 9. At 3CD, the DOP was 25.0 cm, with a crater diameter of 6.75 cm. At 4CD, the DOP was 24.05 cm, with a crater diameter of 8.1 cm. Finally, at 5CD, the DOP was 21.83 cm, with a crater diameter of 5.31 cm. Although the variations were subtle, a distinct trend was observed where the DOP peaked at 2CD and then began to decline. This pattern suggests that at very short standoff distances, the jet does not fully form before impacting the target, diminishing its penetrative capability. Conversely, at distances beyond the optimal point, the jet's effectiveness gradually decreases.

**Table 10** Computational efficiency of the SPH simulation

| Simulation         | Number of particles | CPU computing time | GPU computing time | Speedup |
|--------------------|---------------------|--------------------|--------------------|---------|
| Al-Al HVI          | ~ 450,000           | 8 h                | 20 min             | × 24    |
| Al-Cu HVI          | ~ 120,000           | 6 h                | 10 min             | × 36    |
| Al-concrete HVI    | ~ 1,700,000         | 35 h               | 1 h                | × 35    |
| SC jet penetration | ~ 20,000,000        | X                  | 72 h               |         |

## 5 Conclusions

In this study, we developed a GPU-parallelized SPH solver to simulate extreme collisions involving metal and concrete materials. This advanced solver incorporates multiphase SPH models, complex constitutive models, and EOS, all optimized with variable smoothing lengths to enhance accuracy. To assess the capability and precision of our solver, we conducted numerical simulations of HVI on both metal and concrete materials. Shock waves resulting from impacts were confirmed to propagate well according to material properties, as evidenced by the stable pressure fields that could be observed. Furthermore, it was possible to qualitatively verify that the subsequent formation of cracks exhibited a logically coherent and realistic progression. Quantitatively, comparative analysis with experimental observations demonstrated that our solver offers improved accuracy over previous approaches, adeptly capturing large deformations and the propagation of fractures. In addition, the CUDA-based implementation resulted in dramatic computational efficiency—up to  $35\times$  faster than an optimized multi-core CPU build—enabling 20-million-particle simulations that were otherwise infeasible as seen in Table 10.

Based on well-proven benchmarks involving metal and concrete materials, used respectively in shaped charge liners and as target materials, collisions between shaped charge jets and concrete targets were simulated. It was successfully replicated in the numerical experiment that the copper liner transformed into a jet due to the explosion, and the dynamic response results for the shock and cracks generated upon impacting the concrete were observed. The relative errors for the simulated DOP and average borehole diameter compared to experimental results were under 5%. Notably, even for the crater diameter, which previously exhibited larger errors, the error was greatly reduced in comparison to previous studies. Furthermore, parametric tests were conducted to examine variations in the geometry of the shaped charge and the compressive strength of the concrete. These tests demonstrated reasonable variations in the morphology of the crater, including the DOP and borehole diameter. Consequently, it was demonstrated that our SPH solver is capable of addressing a wide range of shaped charge collision problems. This indicates that SPH simulation has the potential to

serve as a viable and promising alternative to experiments with extreme conditions.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Competing interests** The authors declare no competing interests.

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